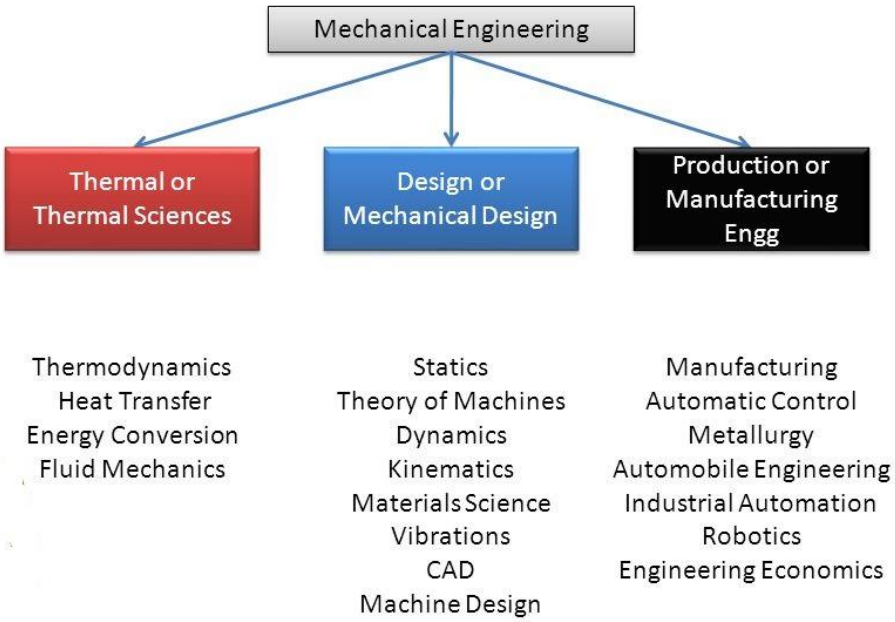
UNIT 04 - Introduction to Mechanical Engineering, Introduction to laws of Thermodynamics, IC Engines, Power Plant, Introduction to Automobiles, Quality assurance and quality control.

Mechanical Engineering is the branch of engineering concerned with the design, manufacture, installation, and operation of engines and machines and with manufacturing processes. It is particularly concerned with forces and motion. Mechanical engineering is one of the oldest and broadest engineering disciplines. It requires a solid understanding of core concepts including mechanics, kinematics, thermodynamics, fluid mechanics, and energy.

The scope of mechanical engineering includes the design and manufacturing of industrial machinery, the best state-of-the-art motor vehicles, and aircraft. Moreover, the maintenance and testing of such machinery are key responsibilities of mechanical engineers.

Mechanical engineer work on teams to develop a wide range of products and systems including, transmissions, engine parts, aircraft engines, control systems, prosthetic devices, disk drives, printers, semiconductor tools, sensors, gas turbines, wind turbines, fuel cells, compressors, robots, machine tools etc. Mechanical Engineer designs, develops, prototypes, and tests all aspects of mechanical components, equipment, and machinery. Develops specifications for new parts or prototypes.

Branches of Mechanical Engineering.



Thermodynamics is said to be a branch of physics that deals with heat, work, and temperature and their relation with the energy and radiation and physical properties of matter.

System - A thermodynamic system is a specific portion of matter with a definite boundary on which our attention is focused. The system boundary may be real or imaginary, fixed or deformable.

Surrounding - Everything outside the system that has a direct influence on the behaviour of the system is known as a surrounding.

There are three types of systems:
(i) Open System: It exchanges both energy and matter with the surrounding.
(ii) Closed System: It exchanges only energy but not matter with surroundings.
(iii) Isolated System: It exchanges neither energy nor matter with the surrounding.

Enthalpy is the measurement of energy in a thermodynamic system. The quantity of enthalpy equals the total heat content of a system, equivalent to the system's internal energy plus the product of volume and pressure.
Mathematically, the enthalpy, H, equals the sum of the internal energy, E, and the product of the pressure, P, and volume, V, of the system.

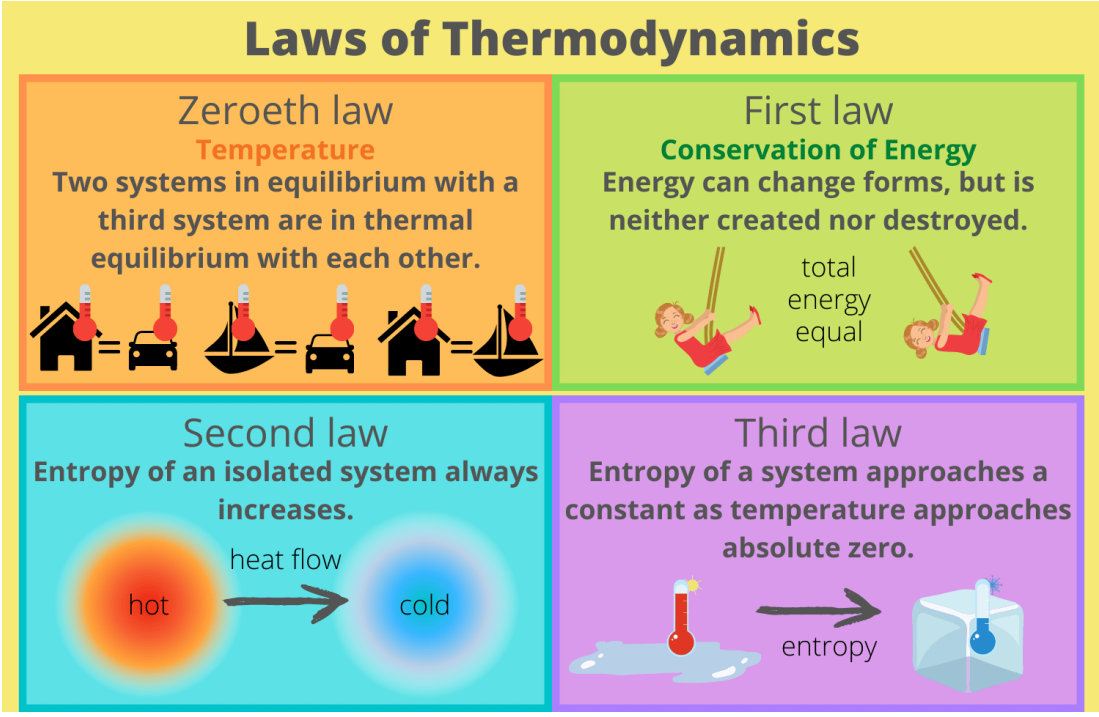
$$H = E + PV$$

Entropy is a thermodynamic quantity whose value depends on the physical state or condition of a system. In other words, it is a thermodynamic function used to measure the randomness or disorder.
For example, the entropy of a solid, where the particles are not free to move, is less than the entropy of a gas, where the particles will fill the container.

Laws of Thermodynamics -

There are four laws of thermodynamics and are given below:

- Zeroth Law of Thermodynamics
- First Law of Thermodynamics
- Second Law of Thermodynamics
- Third Law of Thermodynamics



Zeroth Law of Thermodynamics

The Zeroth law of thermodynamics states that if two bodies are individually in equilibrium with a separate third body, then the first two bodies are also in thermal equilibrium with each other.

This means that if system A is in thermal equilibrium with system C and system B is also in equilibrium with system C, then system A and B are also in thermal equilibrium.

First Law of Thermodynamics

First law of thermodynamics, also known as the law of conservation of energy, states that energy can neither be created nor destroyed, but it can be changed from one form to another.
The first law of thermodynamics may seem abstract, but we will get a clearer idea if we look at a few examples of the first law of thermodynamics.

Examples:

- Plants convert the radiant energy of sunlight to chemical energy through photosynthesis. We eat plants and convert the chemical energy into kinetic energy while we swim, walk, breathe, and scroll through this page.
- Switching on light may seem to produce energy, but it is electrical energy that is converted.

Second Law of Thermodynamics

Second law of thermodynamics states that the entropy in an isolated system always increases. Any isolated system spontaneously evolves towards thermal equilibrium—the state of maximum entropy of the system. The entropy of the universe only increases and never decreases.

Third Law of Thermodynamics

Third law of thermodynamics states that the entropy of a system approaches a constant value as the temperature approaches absolute zero.
The entropy of a pure crystalline substance (perfect order) at absolute zero temperature is zero. This statement holds true if the perfect crystal has only one state with minimum energy.

Example -

1. The molecules within it move freely and have high entropy.
2. If one decreases the temperature below 100 °C, the steam gets converted to water, where the movement of molecules is restricted, decreasing the entropy of water.
3. When water is further cooled below 0 °C, it gets converted to solid ice. In this state, the movement of molecules is further restricted and the entropy of the system reduces more.
4. As the temperature of the ice further reduces, the movement of the molecules in them is restricted further and the entropy of the substance goes on decreasing.
5. When the ice is cooled to absolute zero, ideally, the entropy should be zero. But in reality, it is impossible to cool any substance to zero.

Heat engine:

A heat engine is a device which transforms the chemical energy of a fuel into thermal energy and uses this energy to produce mechanical work.
It is classified into two types-

(a) External combustion engine

(b) Internal combustion engine

External combustion engine:

In this engine, the products of combustion of air and fuel transfer heat to a second fluid which is the working fluid of the cycle.

Examples:

- In the steam engine or a steam turbine plant, the heat of combustion is employed to generate steam which is used in a piston engine (reciprocating type engine) or a turbine (rotary type engine) for useful work.
- In a closed cycle gas turbine, the heat of combustion in an external furnace is transferred to gas, usually air which the working fluid of the cycle.

Internal combustion engine:

In this engine, the combustion of air and fuels take place inside the cylinder and are used as the direct motive force.

It can be classified into the following types:

1. According to the basic engine design -

(a) Reciprocating engine (Use of cylinder piston arrangement), (b) Rotary engine (Use of turbine)

2. According to the type of fuel used -

(a) Petrol engine, (b) diesel engine, (c) gas engine (CNG, LPG), (d) Alcohol engine (ethanol, methanol etc)

3. According to the number of strokes per cycle -

(a) Four stroke and (b) Two stroke engine

4. According to the method of igniting the fuel -

(a) Spark ignition engine, (b) compression ignition engine and (c) hot spot ignition engine

5. According to the working cycle -

(a) Otto cycle (constant volume cycle) engine, (b) diesel cycle (constant pressure cycle) engine, (c) dual combustion cycle (semi diesel cycle) engine.

6. According to the fuel supply and mixture preparation -

(a) Carburetted type (fuel supplied through the carburettor), (b) Injection type (fuel injected into inlet ports or inlet manifold, fuel injected into the cylinder just before ignition).

7. According to the number of cylinders -
(a) Single cylinder and (b) multi-cylinder engine

8. Method of cooling - water cooled or air cooled

9. Speed of the engine - Slow speed, medium speed and high speed engine

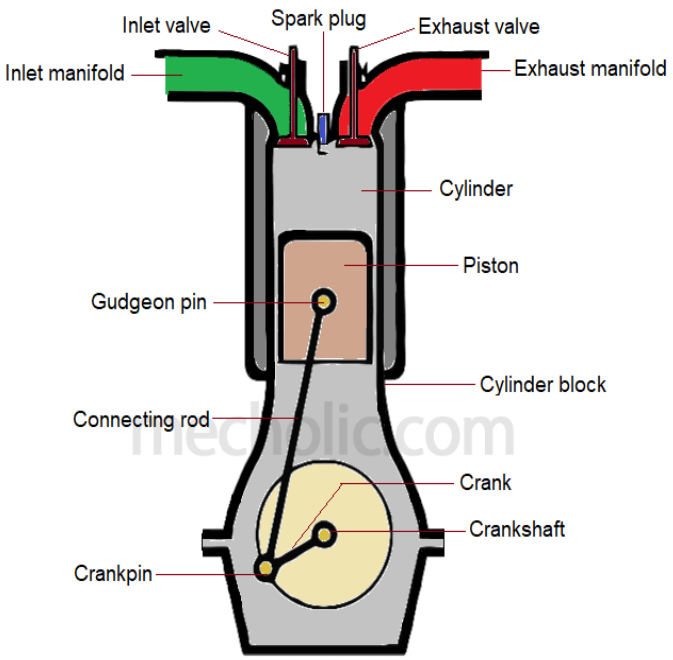
10. Cylinder arrangement -Vertical, horizontal, inline, V-type, radial, opposed cylinder or piston engines.

11. Valve or port design and location - Overhead (I head), side valve (L head); in two stroke engines: cross scavenging, loop scavenging, uniflow scavenging.

12. Method governing - Hit and miss governed engines, quantitatively governed engines and qualitatively governed engine

13. Applications - Automotive engines for land transport, marine engines for propulsion of ships, aircraft engines for aircraft propulsion, industrial engines, prime movers for electrical generators.

Main components of reciprocating IC engines:



Cylinder: It is the main part of the engine inside which piston reciprocates to and fro. It should have high strength to withstand high pressure above 50 bar and temperature above 2000 oC. The ordinary engine is made of cast iron and heavy duty engines are made of steel alloys or aluminum alloys. In the multi-cylinder engine, the cylinders are cast in one block known as cylinder block.

Cylinder head: The top end of the cylinder is covered by cylinder head over which inlet and exhaust valve, spark plug or injectors are mounted. A copper or asbestos gasket is provided between the engine cylinder and cylinder head to make an air tight joint.

Piston: Transmit the force exerted by the burning of charge to the connecting rod. Usually made of aluminium alloy which has good heat conducting property and greater strength at higher temperature.

Piston rings: These are housed in the circumferential grooves provided on the outer surface of the piston and made of steel alloys which retain elastic properties even at high temperature. 2 types of rings- compression and oil rings. Compression ring is upper ring of the piston which provides air tight seal to prevent leakage of the burnt gases into the lower portion. Oil ring is lower ring which provides effective seal to prevent leakage of the oil into the engine cylinder.

Connecting rod: It converts reciprocating motion of the piston into circular motion of the crank shaft, in the working stroke. The smaller end of the connecting rod is connected with the piston by gudgeon pin and bigger end of the connecting rod is connected with the crank with crank pin. The special steel alloys or aluminium alloys are used for the manufacture of connecting rod.

Crankshaft: It converts the reciprocating motion of the piston into the rotary motion with the help of connecting rod. The special steel alloys are used for the manufacturing of the crankshaft. It consists of eccentric portion called crank.

Crank case: It houses cylinder and crankshaft of the IC engine and also serves as sump for the lubricating oil.

Flywheel: It is big wheel mounted on the crankshaft, whose function is to maintain its speed constant. It is done by storing excess energy during the power stroke, which is returned during other stroke.

A two-stroke engine is an engine that completes a power cycle with two strokes of the piston during only one crankshaft revolution.

A four-stroke engine is an internal combustion engine in which the piston completes four separate strokes while turning the crankshaft.

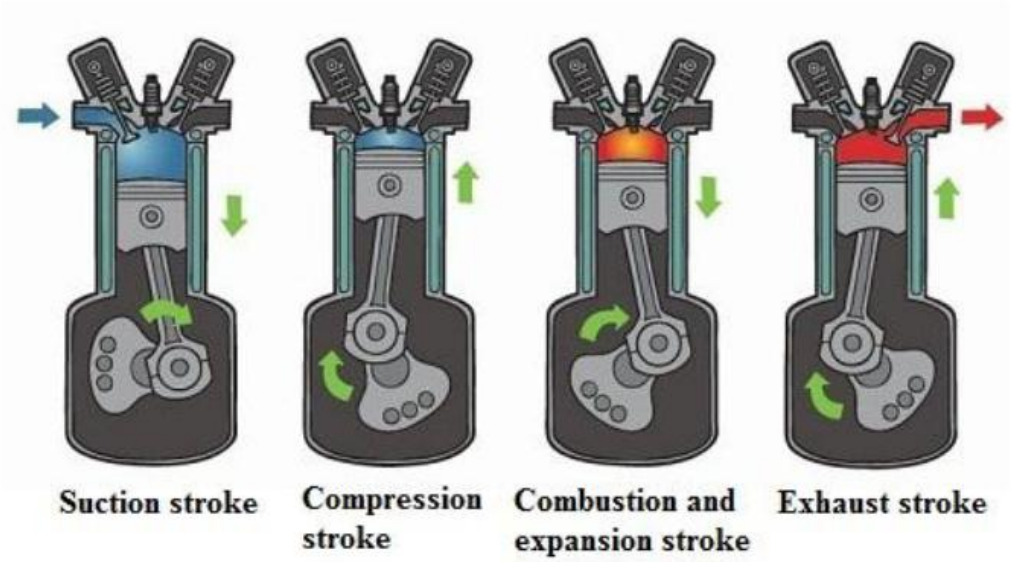


Fig. 2. Cycle of operation in four stroke engine

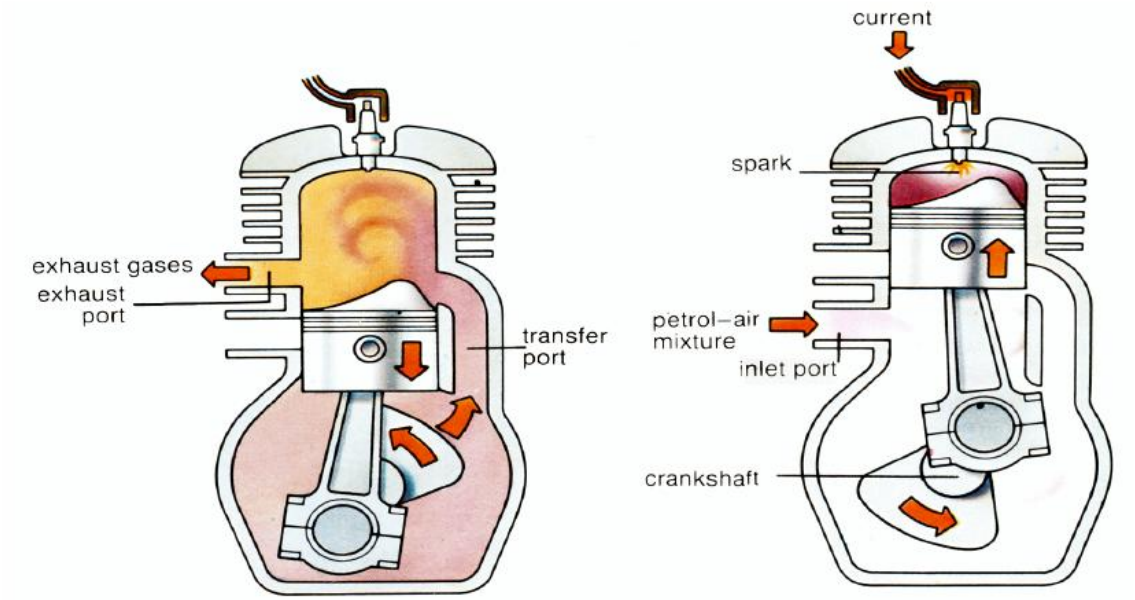


Fig. 3. Cycle of operation in two stroke engine

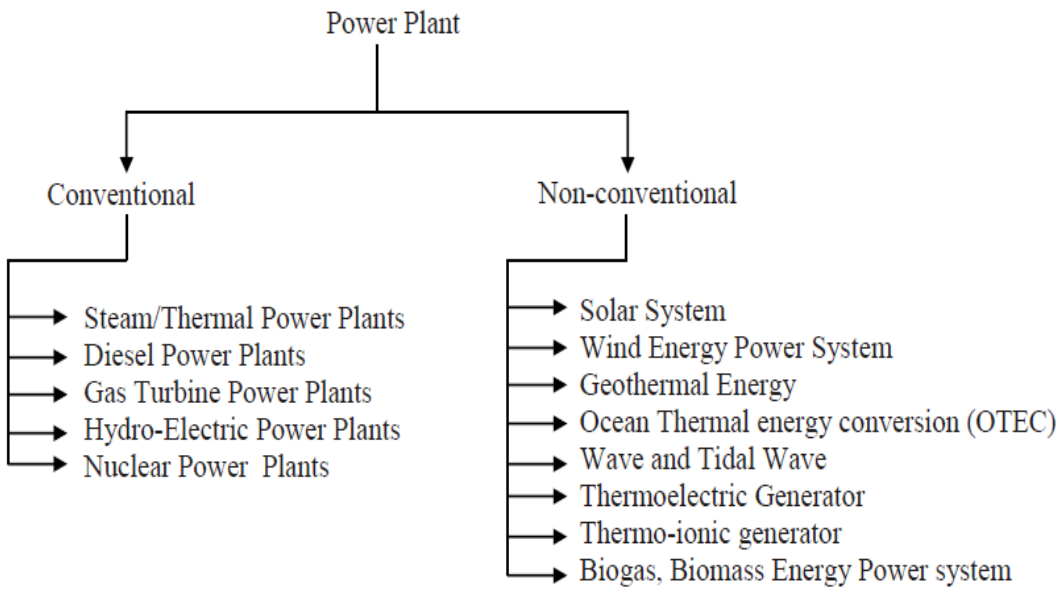
Comparison of Four-stroke and two-stroke engine:

	Two Strokes	Four Strokes
1	It has one revolution of the crankshaft during one power stroke.	It has two revolutions of the crankshaft during one power stroke.
2	It generates high torque.	It generates less torque.
3	It uses a port for the fuel's outlet and inlet.	It uses valves for the fuel's outlet and inlet.
4	Its engines result in lesser thermal efficiency.	Its engines result in higher thermal efficiency.
5	It has a larger ratio in terms of power to weight.	It has a lesser ratio in terms of power to weight.
6	It generates more smoke and shows less efficiency.	It generates less smoke and shows more efficiency.
7	Requires more lubricating oil as some oil burns with the fuel.	Requires less lubricating oil.
8	Due to poor lubrication, more wear and tear occurs.	Less wear and tear occurs.
9	Engines are cheaper and are simple to manufacture.	Engines are expensive due to lubrication and valves and are tough to manufacture.
10	Engines are basically lighter and noisier.	Engines are basically heavier because their flywheel is heavy and less noisy.

Introduction to Power Plant Engineering

A power plant is assembly of systems or subsystems to generate electricity, i.e., power with economy and requirements. The power plant itself must be useful economically and environmental friendly to the society.

CLASSIFICATION OF POWER PLANTS



A power plant may be defined as a machine or assembly of equipment that generates and delivers a flow of mechanical or electrical energy. The main equipment for the generation of electric power is generator. When coupling it to a prime mover runs the generator, the electricity is generated.

The Steam Power Plant, Diesel Power Plant, Gas Turbine Power Plant and Nuclear Power Plants are called **Thermal Power Plant,** because these convert heat into electric energy.

Sources of Energy

There are mainly two types of sources of energy

1. Conventional Sources of Energy (Non-Renewable Sources of Energy)
2. Non-conventional Sources of Energy (Renewable Sources of Energy).

Conventional Sources of Energy

These resources are finite and exhaustible. Once consumed, these sources cannot be replaced by others. Examples include coal, timber, petroleum, lignite, natural gas, fossil fuels, nuclear fuels etc.

The examples are (i) fossil fuel (ii) nuclear energy (iii) hydro energy

Non-Conventional Sources of Energy

These sources are being continuously produced in nature and are not exhaustible. Examples include wood, geothermal energy, wind energy, tidal energy, nuclear fusion, gobar gas, biomass, solar energy etc.

The examples are (i) Solar energy (ii) wind energy (iii) geothermal energy (iv) ocean energy such as tidal energy, wave energy (v) biomass energy such as gobar gas.

Steam Turbine Power Plant

Stage 1
The first conversion of energy takes place in the boiler. Coal is burnt in the boiler furnace to produce heat. Carbon in the coal and Oxygen in the air combine to produce Carbon Dioxide (CO2) and heat.

Stage 2
The second stage is the thermodynamic process :
a) The heat from combustion of the coal boils water in the boiler to produce steam. In modern power plant, boilers produce steam at a high pressure and temperature.

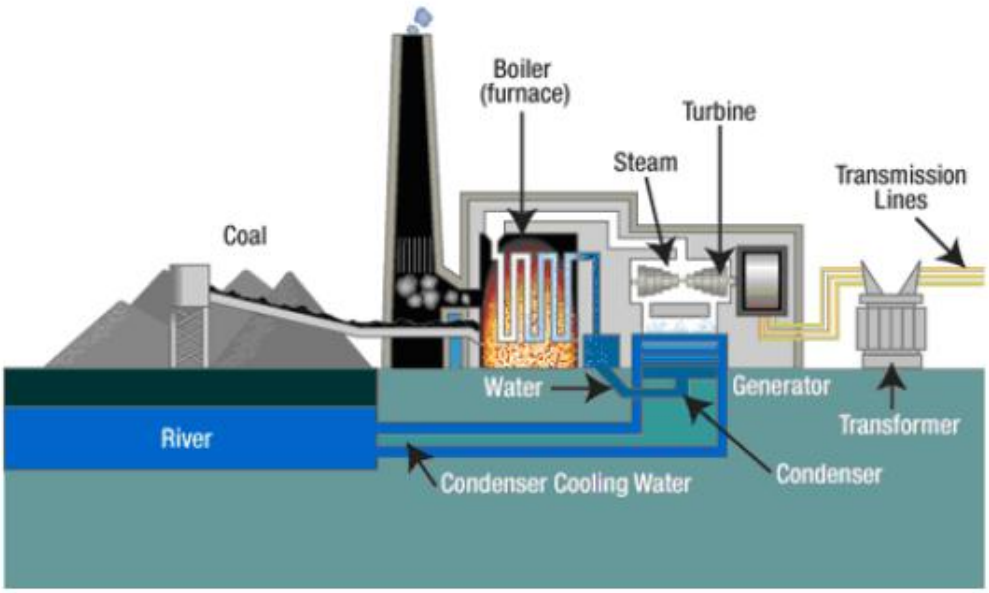
b) The steam is then piped to a turbine.

c) The high pressure steam impinges and expands across a number of sets of blades in the turbine.

d) The impulse and the thrust created rotate the turbine.

e) The steam is then condensed and pumped back into the boiler to repeat the cycle.

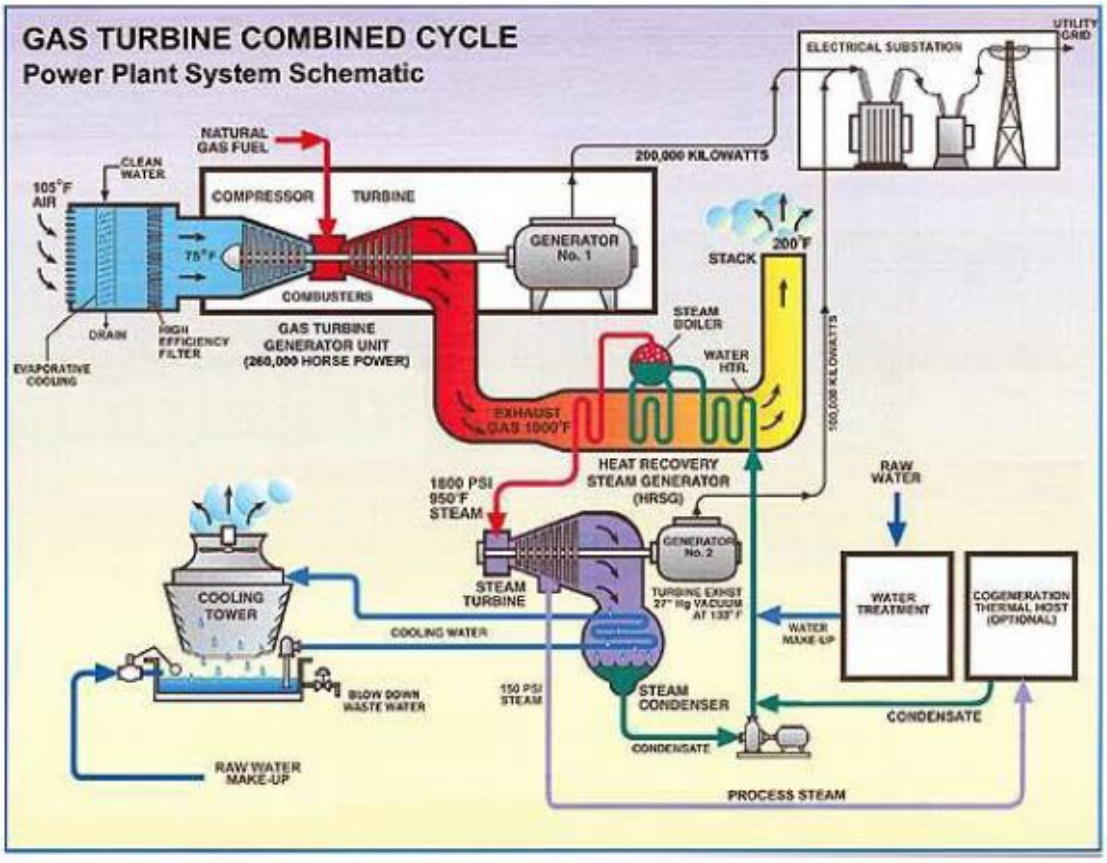
Stage 3
In the third stage, rotation of the turbine rotates the generator rotor to produce electricity based of Faraday's Principle on electromagnetic induction.



Gas Turbine Power Station

The schematic arrangement of a gas turbine power plant is shown in Figure.
The main components of plants are :

- Compressor
- Regenerator
- Combustion Chamber
- Gas Turbine
- Alternator
- Starting motor



Compressor

The compressor used in the plant is generally of rotatory type. The air at atmospheric pressure is drawn by the compressor via the filter which removes the dust from the air. The rotatory blades of the compressor pushthe air between stationary blades to raise its pressure. Thus air at high pressure is available at the output of the compressor.

Regenerator

A regenerator is a device which recovers heat from the exhaust gases of the turbine. The exhaust is passed through the regenerator before wasting to atmosphere. A regenerator consists of a nest of tubes contained in a shell. The compressed air from the compressor passes through the tubes on its way to the combustion chamber. In this way compressor is heated by the hot exhaust gases.

Combustion Chamber The air at high pressure from the compressor is led to the combustion chamber via the regenerator. In the combustion chamber, heat is added to the air by burning oil. The oil is injected through the burner into the chamber at high pressure ensure atomisation of oil and its thorough mixing with air. The result is that the chamber attains a very high temperature. The combustion gases are suitably cooled and then delivered to gas turbine.

Gas Turbine
The products of combustion consisting of a mixture of gases at high temperature and pressure are passed to the gas turbine. These gases in passing over the turbine blades expand and thus do the mechanical work. The temperature of the exhaust gases from the turbine is about 900oF.

Alternator
The gas turbine is coupled into the alternator. The alternator converts the mechanical energy of the turbine into electrical energy. The output of the alternator is given to the bus-bars through transformers, isolators and circuit breakers.

Starting Motor
Before starting the turbine, compressor has to be started. For this purpose, an electric motor is mounted on the same shaft as that of the turbine. The motor is energised by the batteries. Once the unit starts, a part of the mechanical power of the turbine drives the compressor and there is no need of the motor now.

Nuclear Energy

In a universe, energy and matter have a common origin. None the energy nor the matter can be created or destroyed; instead they just change their state. As well, they are convertible to each other. Albert Einstein was the first man who explained this relation by the well known formula : E = mC2
This equation defines :
E (Energy) equals to m (mass) times C2 (C stands for speed of light).

Nuclear Fission

Any try for splitting a part a nucleus will cause a tremendous energy be released. This energy would be released in both forms of heat and light. In a harnessed, controlled way of doing this, a useful energy for producing electricity is possible. Doing this at once would result to a big explosion, as seen in an automatic bomb.

In a nuclear power plant, uranium is the element used as fuel. Uranium is found in many parts of the world but in a low quantity. It is loaded in to the reactor in a tiny pallet form inside long rods. Fission meaning splitting a part is what happens in a reactor. Here uranium atoms are split in a paced controlled chain of reactions.

Nuclear Fusion

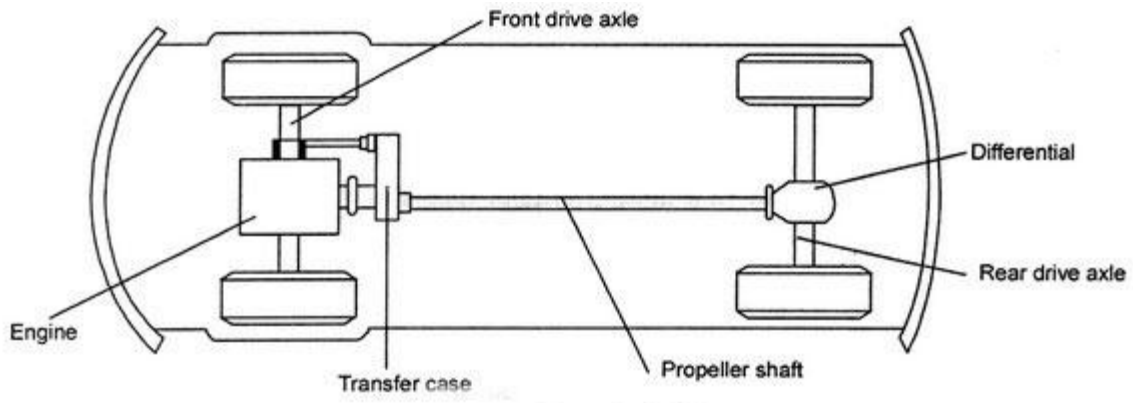
In another form of nuclear reaction, joining of smaller nuclei makes a larger nucleus. Such a process in sun changes the hydrogen atoms to helium. The result heat and light we receive in earth.

In a more detailed explanation, two different types of atoms, deuterium and tritium, combine to make a helium plus and extra particle called neutron.

Automobile engineering is a branch of engineering which deals with everything about automobiles and practices to propel them. Automobile is a vehicle driven by an internal combustion engine and it is used for transportation of passengers and goods on the ground. Examples : Car, jeep, bus, truck, scooter, etc.

The auto components is divided into five segments:

(i) Engine parts
(ii) Drive transmission and steering parts
(iii) Suspension and brake parts
(iv) Electrical parts
(v) Body and chassis



Four wheel drive.

Automobiles or vehicles can be classified on different bases as given below :

On the Basis of Load

- Heavy transport vehicle (HTV) or heavy motor vehicle (HMV), e.g. trucks, buses, etc.
- Light transport vehicle (LTV), e.g. pickup, station wagon, etc.
- Light motor vehicle (LMV), e.g. cars, jeeps, etc.

Wheels

- Two wheeler vehicle, for example : Scooter, motorcycle, scooty, etc.
- Three wheeler vehicle, for example : Autorickshaw, three wheeler scooter for handicaps and tempo, etc.
- Four wheeler vehicle, for example : Car, jeep, trucks, buses, etc.
- Six wheeler vehicle, for example : Big trucks with two gear axles each having four wheels.

Fuel Used

- Petrol vehicle, e.g. motorcycle, scooter, cars, etc.
- Diesel vehicle, e.g. trucks, buses, etc.
- Electric vehicle which use battery to drive.
- Steam vehicle, e.g. an engine which uses steam engine. These engines are now obsolete.
- (e) Gas vehicle, e.g. LPG and CNG vehicles, where LPG is liquefied petroleum gas and CNG is compressed natural gas.

Body - the vehicles are classified as :

(a) Sedan with two doors

(b) Sedan with four doors

(c) Station wagon

(d) Convertible, e.g. jeep, etc.

(e) Van

(f) Special purpose vehicle, e.g. ambulance, milk van, etc.

Transmission

(a) Conventional vehicles with manual transmission, e.g. car with 5 gears.

(b) Semi-automatic

(c) Automatic : In automatic transmission, gears are not required to be changed manually. It is automatically changes as per speed of the automobile.

Position of Engine

Engine in Front Most of the vehicles have engine in the front. Example : most of the cars, buses, trucks in India.

Engine in the Rear Side Very few vehicles have engine located in the rear. Example : Nano car.

Quality means "Performance upon expectations" and "fit for functions." A product is said to be of good quality if it satisfies the customer requirements in terms of performance, grade, durability, appearance and intended use/purpose, etc.

The word "Quality" has a variety of meanings and definitions which are mentioned below:

Fitness for use/purpose: The component is said to possess good quality if it works well in the equipment for which it is meant. In other words, the product should be suitable for the intended use/purpose. Thus it is defined as fitness for purpose.

For example, a gear utilized in a sugarcane juice extracting machine might not possess a good quality surface finish, tolerance, and accuracy as compared with the gear utilized in the top stock of a lathe.

Conformance to requirements: It is the ability of the material/component to perform satisfactorily in the application for which it is intended by the user. Quality of a product, thus, means conformance to customer's requirements. This is a **Manufacturing-based** definition.

Grade: It is a distinguishing feature or grade of the product in appearance, performance, life, reliability, taste, and maintainability, etc. This is generally called **quality characteristics.**

Degree of Excellence: It is a measure of a degree of excellence at a suitable price and control of variability at a suitable cost. this is often a Value-based definition.

Degree of preference: It is the degree to which a particular product is preferred over competing products of similar grade, supported comparative test by customers, normally called as customer's preference.

Quality assurance and quality control are two aspects of quality management. While some quality assurance and quality control activities are interrelated, the two are defined differently. Typically, QA activities and responsibilities cover virtually all of the quality system in one fashion or another, while QC is a subset of the QA activities. Also, elements in the quality system might not be specifically covered by QA/QC activities and responsibilities but may involve QA and QC.

Quality Assurance

QA contains all those planned and systematic actions necessary to supply adequate confidence that a product or service will satisfy/fulfill given requirements for Quality. QA includes QC, but it emphasizes quality in the design of products and processes.

Quality assurance can be defined as "part of quality management focused on providing confidence that quality requirements will be fulfilled." The confidence provided by quality assurance is twofold—internally to management and externally to customers, government agencies, regulators, certifiers, and third parties. An alternate definition is "all the planned and systematic activities implemented within the quality system that can be demonstrated to provide confidence that a product or service will fulfill requirements for quality."

Quality Control

QC is the process through which we measure the actual quality performance and compare it with the set standards. And if there is a deviation then we take corrective actions. The typically quality control program is based on the periodic inspection, followed by feedback of the results and change or adjustment whenever required.

Quality control can be defined as "part of quality management focused on fulfilling quality requirements." While quality assurance relates to how a process is performed or how a product is made, quality control is more the inspection aspect of quality management. An alternate definition is "the operational techniques and activities used to fulfill requirements for quality."